# 4.1 Translations

Name _____ Period ____ Date _____

Geometry 4.1 · Student sheet

**I can…**

- name a vector and its components.
- translate a figure using a vector or a coordinate rule.
- compose translations and write one rule for them.

**Vocabulary** · fill in the missing word

**Vector** — a quantity that has both _____ and magnitude, shown as an arrow

**Transformation** — a _____ that moves or changes a figure in some way

**Preimage and image** — the original figure and the new figure after a _____

**Translation** — a transformation that moves every point of a figure the same distance in the same _____

**Rigid motion** — a _____ that preserves length and angle measure

## Part 1 · Discovery: slide it

Trace △ABC with A(−4, 1), B(−1, 3), C(−2, −1). Slide it 5 right and 2 down.

| Point | Before | After | Change in x | Change in y |
|---|---|---|---|---|
| A | (−4, 1) | | | |
| B | (−1, 3) | | | |
| C | (−2, −1) | | | |

Rule: (x, y) → ( _____ , _____ )

## Part 2 · Notes and examples (fill in each note, then work the example)

**Vector notation:** ⟨a, b⟩ means move a units horizontally and b units _____.

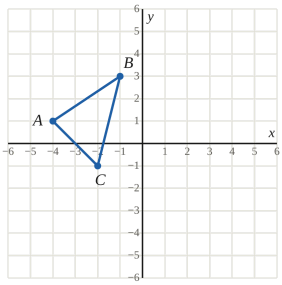
### Example 1

Name the vector from A(1, 2) to B(4, −1) in component form.

**Coordinate rule for a translation:** (x, y) → _____

### Example 2

Translate △ABC by the vector ⟨5, −2⟩. Give the image's vertices.



### Example 3

A translation maps (2, 5) to (−1, 9). Write the rule.

**Composition Theorem:** the composition of two (or more) rigid motions is a _____ motion.

### Example 4

Translate by $(x, y) \to (x + 2, y)$, then by $(x, y) \to (x, y - 4)$. Write one rule, and find the image of (1, 1).

### Example 5

A robot moves from (0, 0) to (3, 2), then from (3, 2) to (1, 7). What single vector takes it from start to finish?

## Part 3 · Partner practice

### Partner 1

Translate P(−2, 6) by $\langle 4, -7 \rangle$.

### Partner 2

A translation maps (3, −1) to (−2, 4). Write the rule.